Mutilple Hypothesis Testing

MFE5340 - AI in FE: Quantitative Investment

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1 The Problem

Multiple hypothesis testing refers to the simultaneous testing of several null hypotheses using the same dataset. In empirical asset pricing, mining hundreds or thousands of factors from the same historical data exemplifies multiple hypothesis testing. When multiple hypotheses are tested simultaneously, false positive appears simply due to luck (noise).

To understand this, let's revisit an important concept in hypothesis testing: the p-value. It represents the conditional probability of observing the test statistic or a more extreme value, given that the null hypothesis is true. For factors, the null hypothesis typically posits that the expected excess return of the factor is zero, i.e., $\mathcal{H}_0 : \mu = 0$. Using sample data of factor returns, we can construct the test statistic (t-statistic):

$$t = \frac{\bar{x}}{s/\sqrt{T}}, \tag{1}$$

where $\bar{x}$ is the time series mean of factor returns, s is the sample standard deviation, and T is the number of periods. The p-value indicates the conditional probability of observing a value more extreme than some cutoff (or threshold) value t_{cutoff} in the student t distribution, assuming $\mathcal{H}_0$ is true (Fisher 1925): p-value $= \text{prob}(t \geq t_{\text{cutoff}}|\mathcal{H}_0)$. (Depending on the specific problem, extreme events can be right-tailed, left-tailed, or two-tailed. Definitions of p-values differ slightly in each case.) For example, a p-value of 0.05 means that the conditional probability of observing a value more extreme than t_{cutoff}, given the null hypothesis, is 5%. The smaller the p-value, the less likely it is to observe t (or $\bar{x}$) under $\mathcal{H}_0$, suggesting a greater tendency to reject the null hypothesis.

In single hypothesis testing, a p-value of 0.05 is often used to determine whether to reject the null hypothesis, corresponding to a t-statistic of approximately 2.0. This cutoff is also known as the significance level. When a null hypothesis is rejected at a certain significance level, the result is said to be significant at that level. However, the presence of multiple hypothesis testing makes a low p-value insufficient to accurately determine the validity of a factor. Consider the scenario where 100 independent factors (i.e., 100 null hypotheses) are tested simultaneously, and a t-statistic of 2.0 is found for one factor. In this case, we cannot claim that the factor is significant at the 0.05 level. Even if all 100 null hypotheses are true (i.e., their excess returns are zero), there is a 99% probability that a t-statistic greater than 2.0 will appear purely by chance. Therefore, using the traditional t-statistic cutoff of 2.0 may lead to many false discoveries (false rejections). These false discoveries, where the null hypothesis is true but wrongly rejected, are statistically known as Type I errors. Thus, correctly addressing the impact of multiple hypothesis testing is crucial for empirical asset pricing and factor investing.

In this regard, academic research can be categorized into two main approaches: frequentist methods and Bayesian methods. To account for luck (noise), frequentist methods aim to control Type I errors by increasing the standard errors, thereby lowering the significance level of individual factors. Intuitively, increasing the standard errors means expanding the confidence intervals, thus making the significance cutoff for individual tests more stringent. As a result, only factors with

original t-statistics far exceeding the traditional level of 2.0 are likely to remain significant after the adjustment. Recent empirical studies have indicated that the t-statistic cutoff for assessing factor significance has been raised to above 3.4, well beyond 2.0. The other approach is the Bayesian method, which aims to calculate the *posterior* probability of a factor being true. Its advantage lies in the ability to incorporate appropriate financial priors within the Bayesian framework, but its downside is the complexity of computation, often requiring compromises and simplifications. Besides the standard Bayesian approach, a new extension of Bayesian thinking has emerged, which models the expected excess returns of true and false factors using a bi-modal mean distribution, and pre-determining the proportion of true factors among all factors through prior knowledge. The figure below summarizes the representitative literature for both research streams.

Frequentist	Bayesian
Family-wise error rate	Full Bayesian
• Bonferroni (1936) • Holm (1979) • White (2000) • Romano and Wolf (2005, 2007)	• Scott and Berger (2006) • Harvey, Liu and Zhu (2016)
False discovery rate	Minimum Bayesian factor
• Benjamini and Hochberg (1995) • Benjamini and Yekutieli (2001) • Harvey, Liu and Zhu (2016)	• Harvey (2017)
False discovery proportion	Bayesian extension
• Romano and Wolf (2007) • Romano, Shaikh and Wolf (2008) • Chordia, Goyal and Saretto (2020)	• Harvey and Liu (2020, 2021)

Figure 1: Two Research Streams for Multiple Hypothesis Testing

2 Frequentist Methods

As previously mentioned, the goal of frequentist methods is to control Type I errors. In statistics, family-wise error rate (FWER), the false discovery rate (FDR), and the false discovery proportion (FDP) are all Type I error type metrics. These metrics can be explained using Table 1.

Table 1: Contingency Table

	$\mathcal{H}_0$ Accepted	$\mathcal{H}_0$ Rejected	Total
$\mathcal{H}_0$ is True	T_0	F_1	K_0
$\mathcal{H}_0$ is False	F_0	T_1	K_1
Total	$K - L$	$L = F_1 + T_1$	K

Suppose there are K factors being tested, with K_0 true null hypotheses and K_1 false null hypotheses. At a given significance level (e.g., 0.05), L null hypotheses are rejected, including T_1 true discoveries and F_1 false discoveries. We can then define the FWER, FDR, and FDP using F_1 and L.

2.1 Family-Wise Error Rate

The family-wise error rate (FWER) is defined as the probability of at least one false discovery, i.e., $\text{prob}(F_1 \geq 1)$. At a given significance level α_s, the mathematical expression for controlling the FWER is:

$$\text{prob}(F_1 \geq 1) \leq \alpha_s. \tag{2}$$

Controlling the FWER is quite stringent for individual hypotheses, greatly increasing Type II errors. (Type II error refers to the scenario in hypothesis testing where the null hypothesis $\mathcal{H}_0$ is false but is incorrectly accepted.) The earliest algorithms for controlling the FWER include the Bonferroni (1936) correction and the Holm (1979) correction. Both methods adjust the p-values of individual tests to control the FWER. Since the adjusted p-values are typically **larger** than the original p-values (i.e., the adjusted t-statistics are **lower** than the original t-statistics), this reduces the significance of the factors (i.e., many factors are no longer significant). The following hypothetical example illustrates these corrections. Suppose there are six factors with original p-values in ascending order: 0.005, 0.009, 0.0128, 0.0135, 0.045, and 0.06. Without considering the multiple hypothesis testing issue, at a significance level of 0.05, the first five factors would be deemed significant.

Bonferroni (1936) adjustment for the original p-values goes as follows:

$$p_i^{\text{Bonferroni}} = \min\{Kp_i, 1\}, \quad i = 1, \cdots, K, \tag{3}$$

where p_i and $p_i^{\text{Bonferroni}}$ represent the original and adjusted p-values of the i-th factor, respectively, and $K = 6$ indicates the number of factors. According to this definition, the adjusted p-values for these six factors are 0.03, 0.054, 0.0768, 0.081, 0.27, and 0.36. Therefore, after adjustment, only the first factor remains significant at the 0.05 significance level.

Next, let's look at the Holm (1979) adjustment, which adjusts the original p-values sequentially from smallest to largest:

$$p_i^{\text{Holm}} = \min\left\{\max_{j \leq i}\{(K - j + 1)p_j\}, 1\right\}, \quad i = 1, \cdots, K, \tag{4}$$

where p_i^{Holm} is the adjusted p-value for the i-th factor. According to this definition, among the six factors in this example, the adjusted p-value for the factor with the smallest original p-value becomes 0.03 after adjustment. For the factor with the second smallest p-value, the adjusted p-value

is $\max\{6 \times 0.005, 5 \times 0.009\} = 0.045$. Similarly, the adjusted p-values for the other four factors are:

$$\begin{aligned} p_3^{\text{Holm}} &= \max\{6p_1, 5p_2, 4p_3\} = 4p_3 = 0.0512, \\ p_4^{\text{Holm}} &= \max\{6p_1, 5p_2, 4p_3, 3p_4\} = 4p_3 = 0.0512, \\ p_5^{\text{Holm}} &= \max\{6p_1, 5p_2, 4p_3, 3p_4, 2p_5\} = 2p_5 = 0.09, \\ p_6^{\text{Holm}} &= \max\{6p_1, 5p_2, 4p_3, 3p_4, 2p_5, p_6\} = 2p_5 = 0.09. \end{aligned} \tag{5}$$

Thus, after the adjustment, only the first two factors remain significant at the 0.05 significance level.

In recent years, several new algorithms targeting the control of the family-wise error rate have been proposed, including the bootstrap reality check method by White (2000) and the StepM and k-StepM methods by Romano and Wolf (2005, 2007). These three algorithms use the bootstrap method to resample the factor return data and combine it with orthogonalization to determine the cutoff for the t-statistics, thus avoiding any assumptions about the data distribution.

In the descriptions above, the terms *orthogonalization* and *bootstrap* reflect the core of these algorithms and frequentist methods. Orthogonalization eliminates the sample mean of the factor returns, making the factor returns zero-mean random disturbances over time. The bootstrap method samples the orthogonalized returns to derive the distribution of test statistics caused purely by chance, thereby determining whether the significance of the original factors is genuine or merely noise. The recitation will cover these algorithms in more details and apply them on the anomalies in the Chinese stock market. Due to their stringent nature, methods targeting the control of the family-wise error rate are not well-suited for the financial field.

2.2 False Discovery Rate

The false discovery rate (FDR) is defined as the expected value of F_1/L, i.e., $\text{FDR} \equiv \mathbb{E}[F_1/L]$. Controlling it at a given level δ can be expressed as:

$$\mathbb{E}[F_1/L] \leq \delta. \tag{6}$$

From the definition, it is clear that controlling the FDR allows F_1 to increase with L. Since the control target is the proportion of false discoveries, this is a relatively lenient method. This means that when the FDR is used instead of FWER as the control target, more null hypotheses are rejected. Representative algorithms include Benjamini and Hochberg (1995) and Benjamini and Yekutieli (2001). These algorithms are less sensitive to the correlation between test statistics and are more adaptive. Using the six hypothetical factors mentioned earlier, let's illustrate these two adjustment methods (collectively known as the BHY adjustment). The BHY adjustment starts

with the largest original p-value and adjusts in reverse order, as follows:

$$p_i^{\text{BHY}} = \begin{cases} p_K & \text{if} \quad i = K, \\ \min\left\{p_{i+1}^{\text{BHY}}, \dfrac{K \times c(K)}{i} p_i\right\} & \text{if} \quad i \leq K-1, \end{cases} \tag{7}$$

where $c(K) = \sum_{j=1}^{K} \frac{1}{j}$, and p_{i+1}^{BHY} is the adjusted p-value. In this example, since $K = 6$, $c(K) = 2.45$. According to the definition, the adjusted p-value for the factor with the largest original p-value remains unchanged. Then, starting from the second largest, we calculate sequentially according to the formula, and finally obtain the BHY-adjusted p-values for all factors, which are (in ascending order): 0.0496, 0.0496, 0.0496, 0.0496, 0.06, and 0.06. At the 0.05 significance level, the first four factors remain significant. As expected, compared to the FWER adjustments, more factors remain significant when the BHY adjustment is applied.

2.3 False Discovery Proportion

The goal of controlling the false discovery proportion (FDP) is to limit the probability that F_1/L exceeds a given threshold γ to no more than the significance level α_s:

$$\text{prob}(F_1/L \geq \gamma) \leq \alpha_s. \tag{8}$$

From the definition, it is evident that this is similar to the false discovery rate. Common algorithms include Romano and Wolf (2007) and Romano et al. (2008). Algorithms targeting the control of these three statistics are widely applied in statistics. In the financial field, it is generally considered better to target the control of the false discovery rate and the false discovery proportion. Farcomeni (2008), Romano et al. (2008), and Harvey et al. (2020) provide excellent reviews of different algorithms.

In recent years, the academic community has increasingly focused on the impact of multiple hypothesis testing on empirical asset pricing, leading to many excellent research achievements in this area. Below, we briefly review related studies and highlight one of the latest works.

2.4 Recent Development

2.4.1 Harvey, Liu, and Zhu (2016)

One important paper that has sounded the alarm for multiple hypothesis testing in empirical finance is Harvey et al. (2016). This paper studied 316 published factors and aimed to control false discovery rate. It found that only when a factor's original t-statistic exceeds 3.0 does it remain valid after accounting for the impact of multiple hypothesis testing.

Additionally, Harvey et al. (2016) pointed out that the proportion of false discoveries is as high as 27%. Harvey and Liu (2021a) proposed a regression-based framework to control the impact of multiple hypothesis testing. This paper used orthogonalization and the bootstrap method to

derive the empirical distribution of significance levels due to chance, thereby determining the cutoff of significance levels caused by pure luck. If a factor's significance level exceeds this cutoff, it is considered to remain significant after accounting for luck. The advantage of this method is that it can sequentially identify significant factors until no more significant factors are found, helping to evaluate the incremental contribution of each new factor in explaining returns.

When attempting to eliminate the impact of multiple hypothesis testing, another issue that must be faced is how many null hypotheses are being tested simultaneously (i.e., how many factors have been evaluated). This issue is crucial because the base number determines the amount of luck. For example, testing 100 factors versus 10,000 factors makes a significant difference; the best one from the latter is undoubtedly more significant than the best one from the former. Therefore, only by knowing how many factors academia has studied (i.e., both published and unpublished) can the multiple hypothesis testing problem be accurately corrected.

At this point, readers might wonder, given that Harvey et al. (2016) considered over 300 factors and Hou et al. (2020) replicated around 450 factors, are these the total number of factors studied by academia? Unfortunately, the answer is no. These are merely the factors that have been published, and it is unknown how many additional factors were tried behind the scenes. Since the published factors are a subset of all researched factors, we can only infer that the 3.0 cutoff found by Harvey et al. (2016) is conservative. Chordia et al. (2020) used simulation to infer the statistical characteristics of the factor set, and to determine the appropriate cutoff for the t-statistic. Unsurprisingly, this paper further raised the t-statistic cutoff to above 3.4.

2.4.2 Chordia, Goyal, and Saretto (2020)

Let $\mathcal{P}$ represent the set of published factors and $\mathcal{R}$ represent the set of all factors studied. Clearly, $\mathcal{P}$ is a subset of $\mathcal{R}$, and we can reasonably assume the relationship $\mathcal{P} = \{s \in \mathcal{R}, t_s \geq 2.0\}$. Since $\mathcal{P}$ is only a subset (lower bound) of $\mathcal{R}$, using $\mathcal{P}$ to control Type I errors inevitably underestimates the effect of luck.

Now, let's consider the upper bound. Chordia et al. (2020) constructed 2,393,641 factors using variables from the three major financial statements through simple arithmetic operations. Let the set $\mathcal{E}$ represent these factors, forming the upper bound of $\mathcal{R}$. It is worth noting that these factors, derived through simple arithmetic operations, can be viewed as factors found from a purely statistical perspective. So, can $\mathcal{E}$ be used as the researched factor set to control false discoveries? The answer is also no. Unlike statisticians, finance professors with background knowledge are unlikely to use a variable that obviously seems unlikely to predict returns to construct factors. Therefore, the number of factors researched by finance scholars should be (far) fewer than 2,393,641, making $\mathcal{R}$ a subset of $\mathcal{E}$. Combining the above discussions, we get $\mathcal{P} \subset \mathcal{R} \subset \mathcal{E}$.

Despite having found the upper and lower bounds of $\mathcal{R}$, we still do not know how many factors are in the actual $\mathcal{R}$. To solve this problem, Chordia et al. (2020) used simulation. The advantage of simulation is that the data generation process is known, allowing the identification of which $\mathcal{H}_0$

are true and which are false, thereby calculating the correct t-statistic cutoff.

In the simulation, the authors let π be the probability of a significant factor. Additionally, another important variable is Ω, representing the ability of finance scholars to find genuine factors. As mentioned earlier, π is the unconditional probability of a factor. For finance professors, due to their prior knowledge, the conditional probability of discovering a factor is higher than the unconditional probability π, and the parameter Ω is used to calculate this conditional probability, $\Omega\pi$. With this model, the paper calibrated the model parameters using real U.S. stock market data. Once the parameters were calibrated, the paper analyzed factor discovery process using the simulated data generation process, finding that the t-statistic cutoff should be above 3.4. Additionally, they found that the false discovery proportion (i.e., F_1/L) is as high as 45.3%.

3 Bayesian Methods

Frequentist methods rely on introducing metrics that measure the overall Type I error rate for multiple hypotheses (such as family-wise error rate or false discovery rate) and adjust the significance of individual hypothesis tests based on these targets. In contrast, Bayesian methods allow for the incorporation of priors derived from economic and financial theory regarding whether a factor is true. Additionally, Bayesian methods inherently include an Ockham's razor effect, which adjusts the posterior probability of a factor being true based on the number of factors being tested simultaneously.

3.1 Full Bayesian Framework

Scott and Berger (2006) proposed a hierarchical model for studying factor returns within a Bayesian framework. Let X_i represent the average return of factor i. The first hierarchy of this model is the prior distribution of X_i:

$$(X_i|\mu_i, \sigma^2, \gamma_i) \overset{\text{iid}}{\sim} \mathcal{N}(\gamma_i\mu_i, \sigma^2), \tag{9}$$

where μ_i represents the mean, σ^2 represents the variance (the variance is not subscripted by i, implying the assumption that the variances of average returns for all factors are the same), and γ_i is a binary variable, with 1 indicating the factor is genuine and 0 indicating the factor is false. In this model, the assumption of equal variance is not as unreasonable as it sounds. For instance, in reality, one can always adjust leverage to make all factor portfolios equally volatile. However, another critical and stringent assumption of this prior is that the factor returns are conditionally independent and identically distributed. Despite this, conditional independence is crucial for the Bayesian framework and the construction of the likelihood function.

Next, let $\mathbf{X}$, $\boldsymbol{\mu}$, and $\boldsymbol{\gamma}$ represent the vectors corresponding to X_i, μ_i, and γ_i, respectively. Under

the above assumptions, the likelihood function is:

$$f(\mathbf{X}|\sigma^2, \boldsymbol{\mu}, \boldsymbol{\gamma}) = \prod_i \left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(X_i - \gamma_i \mu_i)^2}{2\sigma^2} \right) \right]. \tag{10}$$

The first hierarchy of the model describes the distribution of the average returns of factors given μ_i and γ_i, but it does not specify how μ_i and γ_i are derived. This is where the second hierarchy of the model comes in, which specifies the priors for μ_i and γ_i:

$$\mu_i|\tau^2 \overset{\text{iid}}{\sim} \mathcal{N}(0, \tau^2), \;\; \gamma_i|p_0 \overset{\text{iid}}{\sim} \text{Ber}(1 - p_0). \tag{11}$$

The above distributions assume that μ_i follows an independent and identically distributed normal distribution (with a mean of zero and a variance of τ^2), and γ_i follows a Bernoulli distribution with parameter p_0, meaning that the probability of $\gamma_i = 0$ is p_0 and the probability of $\gamma_i = 1$ is $1 - p_0$.

Finally, to complete the Bayesian framework, we need a third hierarchy, which specifies the prior distributions for τ, σ, and p_0:

$$(\tau^2, \sigma^2) \sim \pi_1(\tau^2, \sigma^2), \;\; p_0 \sim \pi_2(p_0). \tag{12}$$

One can choose the prior distributions π_1 and π_2 for the parameters based on their preferences. In the absence of sufficient prior knowledge, Scott and Berger (2006) recommend using the following prior distributions:

$$\begin{aligned} \pi_1(\tau^2, \sigma^2) &\propto (\tau^2 + \sigma^2)^{-2}, \\ \pi_2(p_0) &= (1 + \alpha) p_0^{\alpha}. \end{aligned} \tag{13}$$

In π_2, α is a parameter that controls the center of its distribution (for example, when $\alpha = 0$, π_2 becomes a uniform distribution). By definition, the parameter p_0 controls the probability that each factor is false (recall that the probability of $\gamma_i = 0$ is p_0). We may not have enough prior knowledge to accurately depict the distribution of p_0, but common sense suggests that most factors should be false, so the value of p_0 should be close to 1. Additionally, as the number of factors considered simultaneously increases, α can be adjusted based on prior knowledge to further control the distribution of p_0 (making it closer to 1).

Using the Bayesian framework above, one can also calculate the posterior probability of each factor being true. The formula for calculating the posterior probability $p_i \equiv \text{prob}(\gamma_i = 1|\mathbf{X})$ is very complex and is not listed in this write-up. You may refer to the original Scott and Berger (2006) paper for details. From the posterior probability, it can be seen that as the number of hypotheses tested simultaneously (i.e., the number of factors) increases, the posterior probability p_i will approach 0. In other words, as the number of noise signals (false factors) increases, the evidence conveyed by true factors will diminish, reflecting a penalty for multiple hypothesis testing corresponding to frequentist methods. This is the reason why the Bayesian framework inherently includes an Ockham's razor effect, which automatically adjusts the posterior probability of a factor

being true based on the number of factors being tested simultaneously.

Although the complete Bayesian framework is theoretically sound, there are many practical issues. First, as mentioned earlier, its assumptions (especially regarding conditional independence) are too stringent. The second issue is computational: when considering a large number of factors, calculating the posterior probability of each factor being true is extremely challenging. Third, even if we obtain the posterior probability of each factor being true, we still need to establish a decision criterion, i.e., at what cutoff should a factor's posterior probability be considered true. However, there is little guidance on this.

3.2 Minimum Bayes Factor and Posterior p-Value

Given the challenges in practical applications of the full Bayesian framework, researchers have sought alternative ways to leverage Bayesian ideas. In this regard, Harvey (2017) introduced the minimal Bayes factor and used it to calculate the **Bayesian posterior** p-value, which helps determine whether a factor is significant. However, before interpreting this, let's briefly review the background of the Harvey (2017) paper.

This paper was derived from the keynote address by Professor Campbell Harvey, then president of the American Finance Association (AFA), at the AFA annual meeting. With a scholar's scientific attitude and integrity, Professor Harvey critically analyzed the trend in empirical asset pricing research. In pursuit of publishing papers in top finance journals, researchers have excessively focused on pursuing low p-values for factors through various data-snooping techniques, a practice known as p-hacking. Unfortunately, due to intentional or unintentional data manipulation, the use of inadequate statistical tests, misunderstanding of p-values, and ignoring the intrinsic economic logic of factors, many factors created under these pressures fail to hold up in real-world (McLean and Pontiff 2016). Moreover, Smart Beta ETFs, originating from factor investing and now mainstream in the financial industry, also suffer from p-hacking issues. Huang et al. (forthcoming) documented the significant underperformance of these funds after launch compared to their in-sample performance, attributing this to excessive data mining. Both academia and industry have been racing toward low p-values, but have strayed further from scientific rigor.

3.2.1 Meaning of p-value

To help readers better understand the minimal Bayes factor and Bayesian posterior p-value later on, let's first clarify the correct interpretation of p-values. By definition, a p-value represents the conditional probability of observing a certain (extreme) event under the null hypothesis. Therefore, a lower p-value suggests that the average return observed in the sample data is less likely under the null hypothesis (i.e., the expected return of the factor is zero).

If we denote an extreme event by D (e.g., the t-statistic exceeding a certain value), the p-value represents $\text{prob}(D|\mathcal{H}_0)$. However, is "a lower p-value" equivalent to "the factor is more likely to generate excess returns"? For the latter, what we actually care about is the conditional probability

that the null hypothesis is true given the occurrence of the extreme event D, i.e., prob$(\mathcal{H}_0|D)$, but this is not what the p-value answers. Confusing p-value $\equiv$ prob$(D|\mathcal{H}_0)$ with prob$(\mathcal{H}_0|D)$ is simply a mistake!

As a side note, the American Statistical Association has long warned against the misuse of p-values, and many top economics journals (such as *Econometrica* and *American Economic Review*) have included policies discouraging the use of stars, i.e., $\star$, $\star\star$, $\star\star\star$, to indicate statistical significance. In 2019, the official journal of the American Statistical Association, *The American Statistician*, published a special issue titled "Statistical Inference in the 21st Century: A World Beyond $p < 0.05$," with over 40 articles criticizing the misuse of p-values and providing feasible alternatives. In this special issue, the American Statistical Association even directly recommended banning the term *statistically significant*.

3.2.2 Bayes Factor

Let's now discuss how Harvey (2017) used the minimum Bayes factor to calculate the Bayesian posterior probability, thus answering the question that people truly care about: prob$(\mathcal{H}_0|D)$. According to Bayesian statistics, the prior odds ratio, posterior odds ratio, and Bayes factor are related by the following equation:

$$\text{Posterior Odds Ratio} = \text{Prior Odds Ratio} \times \text{Bayes Factor}. \tag{14}$$

In our context, let $\mathcal{H}_0$ and $\mathcal{H}_1$ represent the null hypothesis and alternative hypothesis regarding the expected return of a factor, respectively. The Bayes factor is then defined as the ratio of marginal likelihoods:

$$\text{Bayes Factor} = \frac{p(\text{data}|\mathcal{H}_0)}{p(\text{data}|\mathcal{H}_1)}. \tag{15}$$

Let θ represent the parameter of interest. When testing a factor's risk premium, the null hypothesis is usually that it is zero, so $\mathcal{H}_0$ can be written as $\theta = 0$. Under the simple hypothesis case, we have:

$$p(\text{data}|\mathcal{H}_0) = \int f(\text{data}|\theta)\delta(\theta - 0)d\theta = f(\text{data}|\theta = 0), \tag{16}$$

where δ is the Dirac delta.

However, for the alternative hypothesis, to make the analysis more general, it is often assumed that under $\mathcal{H}_1$, θ follows a prior distribution $\pi_A(\theta)$. In this case, the marginal likelihood function is $\int f(\text{data}|\theta)\pi_A(\theta)d\theta$, so the Bayes factor becomes:

$$\text{Bayes Factor} = \frac{f(\text{data}|\theta = 0)}{\int f(\text{data}|\theta)\pi_A(\theta)d\theta}. \tag{17}$$

For testing factors, the posterior odds ratio is what we are truly concerned with. It tells us the relative posterior probabilities of the null and alternative hypotheses. A low posterior odds ratio indicates a low posterior probability of the null hypothesis, allowing us to safely reject it, meaning

that the factor is likely genuine. However, to calculate the posterior odds ratio, we must first compute the Bayes factor. But as seen from the definition above, calculating it requires specifying the prior distribution under the alternative hypothesis, which is often very difficult. The good news is that among the various possible values of the Bayes factor, there is a special value, known as the minimal Bayes factor (MBF), proposed by Harvey (2017).

3.2.3 Minimum Bayes Factor

To intuitively understand the minimal Bayes factor, let's revisit the equation: posterior odds ratio = prior odds ratio × Bayes factor. This equation can be understood as follows: starting with the prior odds ratio for $\mathcal{H}_0$ and $\mathcal{H}_1$, and then adjusting it by multiplying the Bayes factor to obtain the posterior odds ratio. When the prior odds ratio is given, the smaller the Bayes factor (and thus the lower the posterior odds ratio), the weaker our belief in the null hypothesis becomes in the posterior compared to the prior; conversely, the larger the Bayes factor (and thus the higher the posterior odds ratio), the stronger our belief in the null hypothesis becomes in the posterior compared to the prior. Therefore, the Bayes factor measures the extent to which we deviate from the prior odds ratio after observing sample data, and the minimum Bayes factor provides the strongest deviation against the null hypothesis.

With an intuitive understanding of the minimum Bayes factor, we can easily grasp how to calculate it. The MBF corresponds to a special prior distribution under the alternative hypothesis that provides the strongest evidence against the null hypothesis. Consider the following example: suppose there are 1,000 observations of factor returns, with a sample mean of 4%. Under what circumstances would we obtain the MBF? The answer is: when the prior distribution under the alternative hypothesis is concentrated at the sample mean of 4%, i.e., when the density of the prior distribution under the alternative hypothesis is concentrated at the maximum likelihood estimate of the data, the Bayes factor is minimized.

From the above discussion, it is clear that MBF allows us to calculate the lower bound of the posterior probability of the null hypothesis. More importantly, it addresses the question that people truly care about, namely, the conditional probability that the null hypothesis is true given the data. Using the original p-value or t-statistic, Harvey (2017) provided two methods for calculating the MBF:

$$\begin{aligned} \text{MBF} &= -\exp(1) \times p\text{-value} \times \ln(p\text{-value}), \\ \text{MBF} &= \exp(-t\text{-statistic}^2/2). \end{aligned} \tag{18}$$

Furthermore, using the relationship posterior odds ratio = prior odds ratio × Bayes factor and performing some simple algebra, we can easily derive the posterior probability of the null hypothesis being true, which Harvey (2017) referred to as the Bayesian posterior p-value:

$$\text{Bayesian Posterior } p\text{-Value} = \frac{\text{MBF} \times \text{Prior Odds Ratio}}{1 + \text{MBF} \times \text{Prior Odds Ratio}}. \tag{19}$$

To apply the Bayesian posterior p-value in practice, the prior odds ratio must also be specified.

To this end, Harvey provided some heuristic guidelines: (1) for factors with little economic rationale, the prior odds ratio = 49:1; (2) for plausible factors, the prior odds ratio = 4:1; (3) for factors with strong economic theory support, the prior odds ratio = 1:1.

Let's look at a few examples. Head et al. (2009) examined a factor called clever tickers, which posits that certain stock tickers are more appealing to investors than others. The original p-value for this factor was very low, only 0.0079. However, since it lacks strong economic rationale, if we choose a prior odds ratio of 49:1, its Bayesian posterior p-value would be as high as 0.588, indicating that this factor is unlikely to generate excess returns. In the second example, we consider the profitability factor in the Fama and French (2015) five-factor model. Its original p-value was also very low, at 0.0035. Given that this factor has some economic rationale (although it is somewhat difficult to explain from a risk perspective why more profitable stocks should have higher risk premiums), we use a prior odds ratio of 4:1. In this case, the posterior p-value is 0.053, suggesting that it is quitely likely to be a genuine factor. Finally, let's consider the market factor. Its original p-value was 0.01. Since the market factor has strong theoretical backing (originating from the Capital Asset Pricing Model), we use a prior odds ratio of 1:1 and obtain a posterior p-value of 0.035. This shows that the market factor is undoubtedly a genuine factor. These three examples illustrate that, compared to the full Bayesian framework, calculating the Bayesian posterior p-value based on the MBF is more practical.

4 Extension of Bayesian Approach

4.1 Two Issues of Frequentist Approach

This section introduces two recent studies by Harvey and Liu (2020, 2021b). They are not traditional Bayesian methods and they still fall under the category of frequentist methods. However, because they control the proportion of true factors through a prior parameter p_0, they can be seen as extensions of the Bayesian approach. By introducing p_0, these methods not only make the handling of multiple hypothesis testing more realistic but also allow for a trade-off between Type I and Type II errors. (In the context of factor discovering, a Type II error occurs when a factor is truly valid but is incorrectly deemed to have no excess returns.) This trade-off is especially important today as the cost of Type II errors continues to rise.

To recap, the multiple hypothesis testing adjustments in frequentist methods can be summarized as a combined application of orthogonalization and the bootstrap method. Orthogonalization works by eliminating each factor's excess returns within the sample, while the bootstrap method resamples the data after orthogonalization to obtain the distribution of factor return test statistics purely caused by luck. After obtaining this distribution, traditional frequentist methods often select the t-statistic cutoff by controlling a predetermined Type I error rate (such as the 5%) and use this to determine the true factors. However, there are two main issues in this approach:

1. The orthogonalization process implicitly assumes that all factors have zero excess returns,

because it typically orthogonalizes all factors within the sample. In reality, this approach ignores the role of prior information. For many factors under examination, one could reasonably assume, based on financial priors, that a certain proportion of the factors are genuine (such as the market factor). Traditional methods, however, overlook this information.

2. The determination of the t-statistic cutoff generally focuses solely on controlling Type I error without considering the trade-off between Type I and Type II errors. As a result, traditional multiple hypothesis testing methods often have a high Type II error rate, leading to low power (power = 1−Type II error rate). For example, in an extreme case, if an algorithm accepts all null hypotheses, it fails to identify any true factors, resulting in zero power.

As α becomes increasingly scarce (In the context of empirical asset pricing, α represents the portion of expected asset returns that cannot be explained by risk factors), the cost of Type II errors is rising, making the trade-off between the two types of errors more critical. Despite this, traditional methods focus solely on Type I errors (i.e., controlling false discoveries). This is because even for single hypothesis testing, calculating the Type II error rate is not easy, let alone in the context of multiple hypothesis testing. To calculate the Type II error rate, one needs to know the parameter values under the alternative hypothesis. However, for hundreds or thousands of factors, it is impractical to examine their expected excess returns under the alternative hypothesis. This obstacle makes it difficult to apply the methods used to calculate Type II error rates in single tests to multiple hypothesis testing problems.

4.2 Harvey and Liu (2020)

Harvey and Liu (2020) addressed the two issues mentioned above by introducing prior knowledge and using a framework based on a two-stage bootstrap (i.e., double-bootstrap) method. For the first issue, they drew inspiration from the bi-modal mean distribution often used in fund research (Barras et al. 2010, Harvey and Liu 2018). Instead of assuming that all funds have zero excess returns (known as the ensemble null prior), the bi-modal mean distribution assumes that the excess returns of all funds come from two distributions: the majority come from a distribution with a mean of zero, while a small portion come from a distribution with a mean greater than zero. Applied to factors, this corresponds to most factors being false, with their expected returns coming from a distribution with a mean of zero, while a small portion of factors are genuine, with their expected returns coming from a distribution with a non-zero mean.

Let $T \times K$ matrix $\mathbf{X}_0$ represent the time series of factor returns, where T is the number of periods and K is the number of factors. This setup is no different from traditional multiple hypothesis testing methods, but Harvey and Liu (2020) differed in that they control the proportion of genuine factors through the parameter p_0 (where p_0 takes a value between 0 and 1), thereby implementing the bi-modal mean distribution. One can choose the value of p_0 based on their own experience (i.e., prior knowledge), which reflects the Bayesian idea. Once p_0 is selected, a natural thought is that the

top p_0K factors with the highest t-statistics among all K factors are genuine. However, since the sample factor return series comes from an unknown population, considering sampling uncertainty, the top p_0K factors among them may not necessarily all be genuine. To address this issue, Harvey and Liu (2020) conducted the first-stage bootstrap sampling.

The first-stage bootstrap sampling is applied to $\mathbf{X}_0$. Each time a bootstrap sample is drawn, a $T \times K$ matrix $\mathbf{X}_i$ is obtained (where the subscript i denotes the i-th bootstrap sample). For $\mathbf{X}_i$, the t-statistic of each factor is calculated, and the top p_0K factors with the highest t-statistics are considered to be genuine for that bootstrap sample. Next, in the original matrix $\mathbf{X}_0$, the average returns of these factors are replaced with their means in $\mathbf{X}_i$, and the returns of the remaining factors in $\mathbf{X}_0$ are demeaned (i.e., treated as random noise), thereby constructing a $T \times K$ matrix $\mathbf{Y}_i$. Through the steps described above, each time $\mathbf{X}_0$ undergoes bootstrap sampling, a new matrix $\mathbf{Y}_i$ is obtained, where p_0K factors have non-zero average excess returns (i.e., they are considered to be genuine), while the remaining $(1-p_0)K$ factors have zero average excess returns (i.e., they are considered to be false). In constructing $\mathbf{Y}_i$, retaining the means of the p_0K factors reflects prior knowledge, while demeaning the returns of the remaining $(1-p_0)K$ factors over time embodies the concept of orthogonalization. This setup uses a single parameter, p_0, to designate the factors for which the null hypothesis should be rejected, thereby bypassing the difficulty of specifying factor return parameters when subsequently calculating the Type II error rate.

The $\mathbf{Y}_i$ matrices obtained from the first-stage bootstrap sampling serve as the original data for the second round of bootstrap sampling, meaning that $\mathbf{Y}_i$ is considered the *population* for the second-stage bootstrap sampling. For each $\mathbf{Y}_i$, as we know which factors are genuine and which are false, the population is known to us. Thus, by performing bootstrap sampling on $\mathbf{Y}_i$, we can construct a contingency table (recall Section 2), making it easy to calculate both Type I and Type II error rates. For the j-th bootstrap sample result of $\mathbf{Y}_i$, we can define the following four variables based on the contingency table:

- $\text{TN}^{i,j}$: The number of null hypotheses correctly accepted (true negative).
- $\text{TP}^{i,j}$: The number of null hypotheses correctly rejected (true positive).
- $\text{FP}^{i,j}$: The number of null hypotheses incorrectly rejected (false positive).
- $\text{FN}^{i,j}$: The number of null hypotheses incorrectly accepted (false negative).

Although we know which factors in $\mathbf{Y}_i$ are genuine or false, to categorize the hypothesis test results of all factors in each bootstrap sample of $\mathbf{Y}_i$ into the four categories above, we still need to specify a t-statistic cutoff. Here's an example to help clarify. Suppose that factor A is significant in $\mathbf{Y}_i$ (i.e., it is one of the p_0K factors). Further, assume that in the j-th bootstrap sample of $\mathbf{Y}_i$, the t-statistic of factor A is 2.0. Now, suppose the pre-selected t-statistic cutoff is 2.3. Since 2.0 ¡ 2.3, the null hypothesis for factor A would be accepted. However, we know that factor A is genuine in $\mathbf{Y}_i$, meaning the null hypothesis is false. But based on this result above, it is incorrectly

accepted, so it would be categorized as $\text{FN}^{i,j}$. This example illustrates that the four variables $\text{TN}^{i,j}$, $\text{TP}^{i,j}$, $\text{FP}^{i,j}$, and $\text{FN}^{i,j}$ are functions of the t-statistic cutoff. This property is a critical aspect of the double bootstrap framework of Harvey and Liu (2020). Using these four variables, the authors further defined the Realized False Discovery Rate (RFDR), the Realized Rate of Miss (RMISS), and the Realized Ratio of False Discoveries to Misses (RRATIO):

$$\text{RFDR}^{i,j} = \begin{cases} \dfrac{\text{FP}^{i,j}}{\text{FP}^{i,j} + \text{TP}^{i,j}}, & \text{if} \quad \text{FP}^{i,j} + \text{TP}^{i,j} > 0, \\ 0, & \text{if} \quad \text{FP}^{i,j} + \text{TP}^{i,j} = 0. \end{cases} \tag{20}$$

$$\text{RMISS}^{i,j} = \begin{cases} \dfrac{\text{FN}^{i,j}}{\text{FN}^{i,j} + \text{TN}^{i,j}}, & \text{if} \quad \text{FN}^{i,j} + \text{TN}^{i,j} > 0, \\ 0, & \text{if} \quad \text{FN}^{i,j} + \text{TN}^{i,j} = 0. \end{cases} \tag{21}$$

$$\text{RRATIO}^{i,j} = \begin{cases} \dfrac{\text{FP}^{i,j}}{\text{FN}^{i,j}}, & \text{if} \quad \text{FN}^{i,j} > 0, \\ 0, & \text{if} \quad \text{FN}^{i,j} = 0. \end{cases} \tag{22}$$

From the definitions, it is easy to see that RFDR corresponds to Type I error; RMISS corresponds to Type II error; and finally, RRATIO is the ratio between the two types of errors. In practice, a total of J bootstrap samples are drawn for each $\mathbf{Y}_i$. Additionally, suppose that I bootstrap samples $\mathbf{Y}_i$ were generated from the original data $\mathbf{X}_0$ in the first-stage bootstrap method. As discussed above, for each of the four variables $\text{TN}^{i,j}$, $\text{TP}^{i,j}$, $\text{FP}^{i,j}$, and $\text{FN}^{i,j}$, the double-bootstrap method ultimately produces $I \times J$ values. Taking the average of these values yields the final Type I error rate (TYPE1), Type II error rate (TYPE2), and the ratio of the two errors (ORATIO):

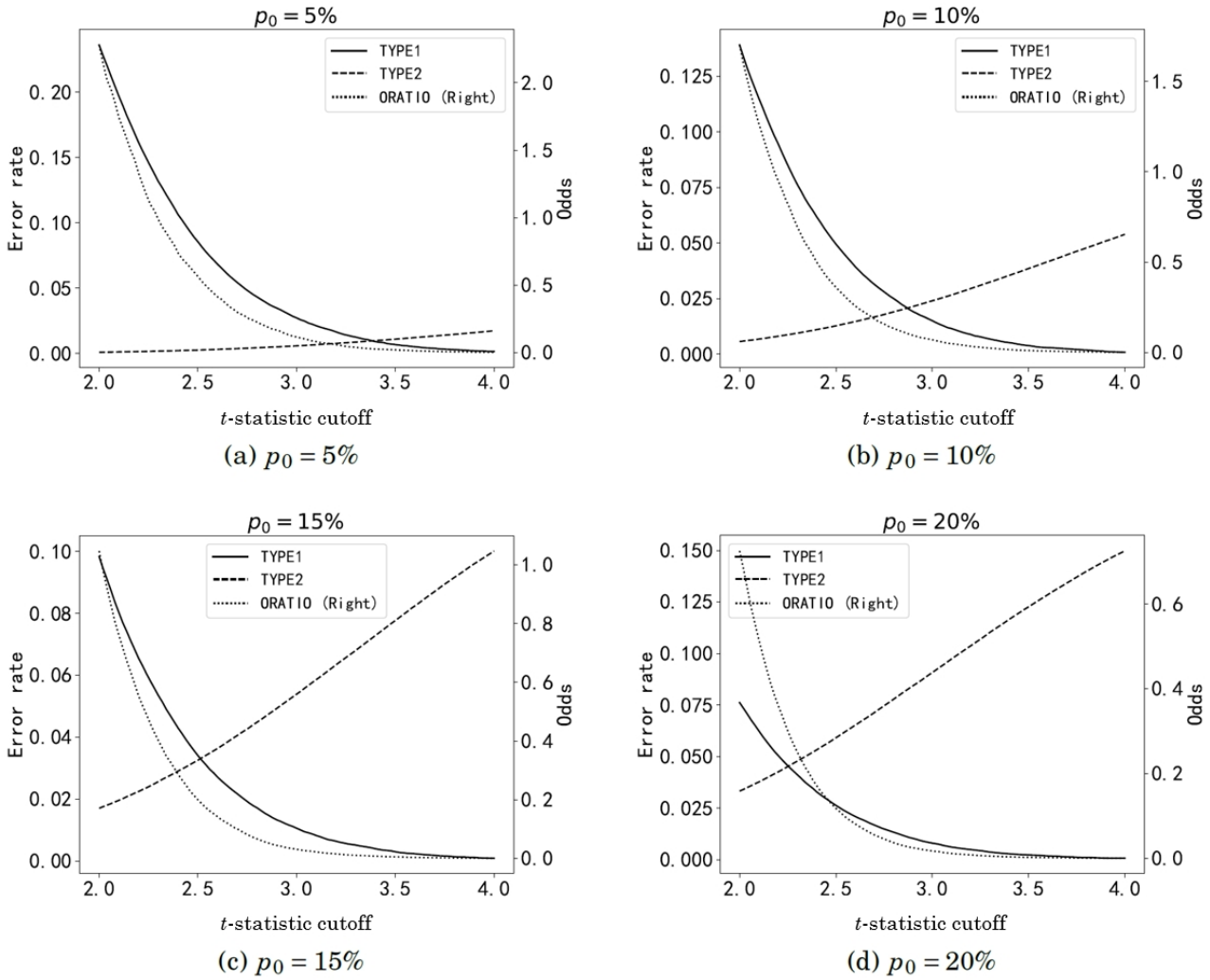
$$\begin{aligned} \text{TYPE1} &= \frac{1}{IJ} \sum_{i=1}^{I} \sum_{j=1}^{J} \text{RFDR}^{i,j}, \\ \text{TYPE2} &= \frac{1}{IJ} \sum_{i=1}^{I} \sum_{j=1}^{J} \text{RMISS}^{i,j}, \\ \text{ORATIO} &= \frac{1}{IJ} \sum_{i=1}^{I} \sum_{j=1}^{J} \text{RRATIO}^{i,j}. \end{aligned} \tag{23}$$

Since $\text{TN}^{i,j}$, $\text{TP}^{i,j}$, $\text{FP}^{i,j}$, and $\text{FN}^{i,j}$ are functions of the t-statistic cutoff, these three final indicators are also functions of the t-statistic cutoff. From the nature of hypothesis testing, we know that TYPE1 decreases as the t-statistic cutoff increases, while TYPE2 increases with the the cutoff, highlighting the trade-off between the two. This allows users to select the most appropriate t-statistic cutoff based on the issue they are most concerned with. For example, if the primary concern is the Type I error rate, then the t-statistic cutoff should be determined by constraining TYPE1 to not exceed a given level (e.g., 5%). If the focus is more on the Type II error rate or the trade-off between the two, the t-statistic cutoff can be chosen by constraining the values of TYPE2 or ORATIO.

Additionally, it is important to note that due to the trade-off between these two types of errors,

when the goal is to control the Type I error rate within a set level, the two-stage bootstrap method ensures that the t-statistic cutoff determined also corresponds to the optimal Type II error rate. In other words, compared to other traditional multiple hypothesis testing methods, the method proposed by Harvey and Liu (2020) has higher power. The authors have demonstrated the superiority of their method through extensive empirical evidence.

The following applies the above two-stage bootstrap method to 95 common factors in the Chinese stock market. The factor return data used in this empirical study comes from *www.factorwar.com*. To save computation time, the empirical study selects $I = 100$ and $J = 100$. To examine the impact of prior knowledge on the results, the empirical study sets the value of p_0 to range from 0% to 20%: $p_0 = 0\%$ means the prior assumes that all factors are false; $p_0 = 20\%$ means the prior assumes that 19 of the factors are genuine. Logically, as p_0 increases, the prior assumes that more factors are genuine, so for a given Type I error rate, the resulting t-statistic cutoff decreases. The empirical results are consistent with these expectations. Figure 4.2 shows the curves of TYPE1, TYPE2, and ORATIO for $p_0 = 5\%$, 10%, 15%, and 20%. The horizontal axis represents the t-statistic cutoff.

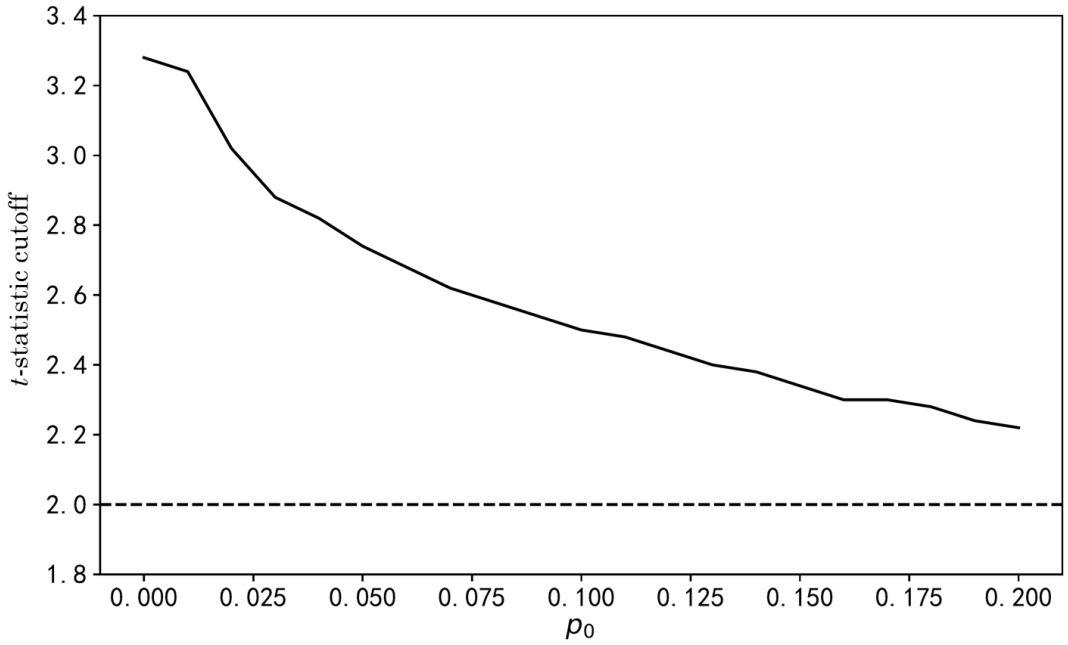


(a) $p_0 = 5\%$ (b) $p_0 = 10\%$

(c) $p_0 = 15\%$ (d) $p_0 = 20\%$

When p_0 is high (i.e., when our expectation of the proportion of genuine factors among those being tested is high), the Type I error rate decreases rapidly as the t-statistic cutoff increases, while the Type II error rate climbs quickly. This is consistent with expectations because when

p_0 is high, if the chosen t-statistic cutoff is too high (i.e., too stringent), it is likely that genuine factors will be missed, resulting in a Type II error. Figure 4.2 shows the t-statistic cutoff (vertical axis) required to control the Type I error rate at 5% for different values of p_0. As p_0 increases, the cutoff decreases monotonically. This figure demonstrates the importance of Bayesian thinking. In traditional methods, since p_0 is not specified, orthogonalization is applied to all factors, resulting in a t-statistic cutoff that is too high (corresponding to the case where $p_0 = 0$). However, when there is sufficient reason to provide a reasonable prior for the factors being tested, an appropriate p_0 can yield a more realistic cutoff, minimizing the Type II error rate at a given Type I error rate level.



4.3 Criticism and Response

In recent years, Professor Harvey and his collaborators have been actively calling on the academic community to resist the unhealthy trend of pursuing extremely low p-values. Their numerous empirical results not only challenge a significant portion of the factors unearthed in academic research over the past few decades but also, to some extent, question the authority of academic research itself. However, to protect the integrity of academic work and research outcomes, voices of dissent have emerged from the other side of the debate as well. (After all, the academic community (especially top-tier journals) is not inclined to admit that many empirical asset pricing articles published in prestigious journals may contain false discoveries.) Among these, the critiques by Chen (2021) and Jensen et al. (2023) are particularly notable. (Both of these articles were published in *Journal of Finance*, one of the top journals in finance, and the official publication of the American Finance Association. This reflects the mainstream academic community's skepticism regarding the existence of (a large number of) false discoveries.)

Chen (2021) conducted a thought experiment and pointed out that p-hacking alone cannot explain the many highly significant factors discovered in academia (such as those with t-statistics

exceeding 6.0 or 8.0). Using his model, he derived a series of conclusions, indirectly suggesting that the concerns about p-hacking might have been exaggerated by Professor Harvey and his collaborators. However, both the academic and professional communities agree that the null hypothesis that all factors have zero expected returns (i.e., the ensemble null prior) will certainly be rejected, meaning that there is a consensus on the existence of some significant factors. Therefore, no one denies that p-hacking alone cannot explain the discovery of some genuinely significant factors. In fact, Professor Harvey's series of articles never conveyed such a view. However, it is also widely believed that multiple hypothesis testing and publication bias have contributed to the emergence of some false factors. So, among all factors, how many are actually genuine? Furthermore, for these factors identified, what is the shrinkage coefficient of their returns out of sample? Chen (2021) did not address these questions.

The conclusions of Chen (2021) are based on one assumption and one core formula. It uses the ensemble null prior, assuming that the expected returns of all factors are zero. Moreover, each factor has a certain probability of being observed (i.e., published) depending on its t-statistic. With this setup, the paper's most important formula is the expected number of attempted factors:

$$\mathbb{E}[K_0] \geq \frac{\mathbb{E}[\text{num observed } (|t_i| > \bar{t})]}{\text{prob}(|Z| > \bar{t})}, \tag{24}$$

where K_0 represents the number of attempted factors (including both published and unpublished ones), Z is a standard normal random variable, and $\bar{t}$ is the selected t-statistic threshold. This formula indicates the lower bound on the number of factors that need to be explored to observe $\mathbb{E}[\text{num observed } (|t_i| > \bar{t})]$ factors with t-statistics exceeding $\bar{t}$. Using this formula and actual factor return data, Chen (2021) derived a series of implications.

First, among the 210 factors considered in the paper, 26 have t-statistics over 6.0, and the calculations suggest that the academic community would need to investigate at least 13 billion factors to identify 26 factors with such high levels of significance. This implies that even if all the world's finance professors worked nonstop to mine factors, it would take 451 years. Given the impracticality of this conclusion, the paper asserts that p-hacking alone cannot account for such highly significant results. Additionally, the paper concludes that the ratio of unpublished to published factors is as high as $10^{14} : 1$, and under this model, the probability of a factor with a t-statistic of 8.0 being published is 10^{10} times higher than that of a factor with a t-value of 2.5. Faced with such shocking and entirely counterintuitive conclusions, while we can readily accept the idea that "p−hacking alone cannot unearth so many significant factors", we must also ask whether these findings might suggest that the ensemble null hypothesis itself is questionable?

In response to the criticism, Harvey and Liu (2021b) provided a rebuttal. When testing factors, in addition to those that have been published, it is also necessary to consider factors that were abandoned by researchers due to insufficient significance. (As a reminder, this issue was addressed in Chordia et al. (2020), as these constitute the total number of factors attempted.) Harvey and Liu (2021b) continued to use the bi-modal mean prior distribution for factor expected returns,

which assumes that the expected returns of a proportion p_0 of factors are greater than zero. To determine the parameters of the distribution for these factor returns, the authors again utilized the bootstrap method. On the other hand, let K_0 represent the total number of factors studied, with the probability of a factor being published depending on its t-statistic according to the following relationship:

$$\text{prob}(t\text{-statistic}) = \frac{\gamma_1}{2\gamma_1 + \exp(-\gamma_2(t\text{-statistic} - 2.0))}, \tag{25}$$

where γ_1 and γ_2 are positive parameters. This function satisfies the condition that the probability increases monotonically with a factor's t-statistic, and as the t-statistic approaches infinity, the probability approaches 0.5. This can be considered a reasonable assumption because even a very high t-statistic does not guarantee that a factor will be published (for example, if the factor lacks any economic rationale). The parameters $[K_0, p_0, \gamma_1, \gamma_2]$ constitute all the parameters in the Harvey and Liu (2021b) model. Compared to Chen (2021), this model differs in the following important aspects:

1. It considers all factors that have been attempted through K_0, rather than focusing solely on the factors that have been published.

2. It replaces the ensemble null prior with a bi-modal mean prior through p_0, which aligns better with the general understanding of factors.

3. It quantitatively describes the factor publication process using parameters γ_1 and γ_2.

Like Chordia et al. (2020), Harvey and Liu (2021b) constructed a model to simulate the process of factors being published, and the next step was to calibrate the model's parameters using actual data. To obtain the K_0 factors (including both published and unpublished ones), Harvey and Liu (2021b) used the 156 published factors collected by Chen and Zimmermann (2020) and more than 18,000 factors generated through pure data mining in Yan and Zheng (2017) as a combined factor pool.

The first step of the simulation involves drawing K_0 factors from this combined pool. Since published factors must have been tested, each simulation requires that the K_0 factors include the 156 published factors; the remaining $K_0 - 156$ factors are randomly selected from the pool of purely data-mined factors. Let the $T \times 156$ matrix $\mathbf{D}_{CZ}$ represent the return series of the published factors, and the $T \times (K_0 - 156)$ matrix $\mathbf{D}^i_{YZ}$ represent the return series of the remaining factors. Finally, construct the $T \times K_0$ matrix $\mathbf{D}^i = [\mathbf{D}_{CZ}, \mathbf{D}^i_{YZ}]$ as the return series of the K_0 factors generated in the i-th simulation.

The second step of the simulation is to calculate the t-statistics of the factors using a bi-modal mean distribution, where p_0 represents the proportion of genuine factors. First, calculate the absolute values of the t-statistics of all factors using $\mathbf{D}^i$, retain the p_0K_0 factors with the highest absolute values, and mean-adjust the returns of the remaining $(1 - p_0)K_0$ factors over time. The resulting $T \times K_0$ return series matrix is denoted as $\hat{\mathbf{D}}^i$. Considering that sampling uncertainty

can cause discrepancies between the sample distribution and the population distribution, treat $\hat{\mathbf{D}}^i$ as the population and apply the second hierarchy of bootstrap sampling to simulate actual return series. As described earlier, the double-bootstrap method used in Harvey et al. (2020) is employed here. The return series obtained from the j-th bootstrap sample of $\hat{\mathbf{D}}^i$ is denoted as $\hat{\mathbf{D}}^{i,j}$. The third step of the simulation is to determine which factors are published. For each $\hat{\mathbf{D}}^{i,j}$, calculate the t-statistics of all factors and use the formula for prob(t-statistic) to calculate the publication probability p_i. For each factor, a uniformly distributed random variable between 0 and 1 is generated; if its value is less than p_i, the factor is considered published; otherwise, it is not. Let $\tilde{K}_{i,j}$ represent the number of factors ultimately published from bootstrap sample $\hat{\boldsymbol{D}}^{i,j}$, and denote the return series of all published factors as $\tilde{\mathbf{D}}^{i,j}$.

In terms of parameter calibration, for a given set of parameters $[K_0, p_0, \gamma_1, \gamma_2]$, the two-hierarchy bootstrap method can simulate a set of published factors. But which set of parameters is correct? The choice of parameters is based on how closely certain statistical indicators of the simulated published factors match the actual data. To this end, Harvey and Liu (2021b) selected five quantiles (i.e., 5th, 10th, 50th, 90th, and 95th) of the t-statistics of the actual published factors, as well as the number of significant factors, as the calibration's statistical indicators. Since only 132 out of the 156 published factors have t-statistics greater than 2.0, the five quantiles of t-statistic are calculated using these 132 factors, and the number of significant anomalies is 132, not 156. Ultimately, the calibration goal is to select the appropriate parameters to minimize the following objective function, which is the weighted average of the squared errors between the simulated and actual indicators:

$$\text{obj func} = \frac{1}{IJ}\sum_{i=1}^{I}\sum_{j=1}^{J}\left(\sum_{s=1}^{5}\tilde{w}_s\left(\tilde{pc}_s^{i,j} - pc_{0,s}\right)^2 + \tilde{w}_K\left(\tilde{K}_{i,j} - 156\right)^2\right), \tag{26}$$

where $\tilde{pc}_s^{i,j}$ and $pc_{0,s}$ represent the s-th quantile of t-statistic calculated using the bootstrap sample $\tilde{\mathbf{D}}^{n_m}$ and the actual data, respectively. As explained earlier, the simulation uses a two-hierarchy bootstrap method: (1) The first hierarchy bootstrap is performed I times, each time randomly generating K_0 factors and constructing $\hat{\mathbf{D}}^i$; (2) For each $\hat{\mathbf{D}}^i$, the second hierarchy bootstrap is performed J times, each time generating return series through bootstrap sampling and determining which factors are published, resulting in $\tilde{\mathbf{D}}^{i,j}$. Therefore, for any given set of parameters $[K_0, p_0, \gamma_1, \gamma_2]$, there are $I \times J$ simulations, and the objective function is the mean of their errors. Since multiple simulations were conducted, it is easy to calculate the standard deviation of each statistical indicator. In the objective function, $\tilde{w}_s$ and $\tilde{w}_K$ are equal to the reciprocal of the standard deviation, determining the weight of different statistical indicators in the objective function.

The results of the parameter calibration may be somewhat surprising (but reasonable upon reflection) in that the problem itself is **unidentifiable** (i.e., there is a lack of identification). In other words, the optimal parameters are not unique. The simulated statistics closely match the actual values under the following three sets of parameters:

- $K_0 = 1000$, $p_0 = 0.29$, $\gamma_1 = 1.17$, and $\gamma_2 = 0.45$;

- $K_0 = 2000$, $p_0 = 0.12$, $\gamma_1 = 0.75$, and $\gamma_2 = 0.33$;
- $K_0 = 5000$, $p_0 = 0.06$, $\gamma_1 = 0.21$, and $\gamma_2 = 0.28$.

Among these three sets of parameters, the number of factors studied, K_0, equals 1000, 2000, and 5000, respectively. The reason this problem is unidentifiable is precisely because only the published factors are observed, while the true K_0 (the total number of factors attempted) remains unknown. This issue is inevitable when dealing with p-hacking. As for the possible value of K_0, it depends on the researcher's experience and understanding of the empirical data.

Although the true value of K_0 is unknown, there is indirect evidence that tends to support the third set of parameters (i.e., $K_0 = 5000$). For example, in the first two sets of parameters, the proportion of genuine factors p_0 is 29% and 12%, respectively, both of which are higher than generally assumed (i.e., the proportion of genuine factors relative to all factors should be very low). In the third set of parameters, p_0 is only 6%. On the other hand, if we consider the probability of factors with different t-statistics being published, in the first two sets of parameters, the probability of a factor with a t-statistic of 2.0 being published is over 30%, which seems inconsistent with the increasing difficulty of publishing finance papers in today's competitive academic environment. In the third set of parameters, this probability is halved to less than 15%. Taking the third set of parameters ($K_0 = 5000$) as an example, the model generates the following key conclusions:

1. To control the false discovery rate at 5%, the t-statistic threshold should be 3.0 (this value is consistent with Harvey et al. 2016).
2. Among all **published** factors, the proportion of genuine factors is 62%.
3. Among all published factors, the shrinkage coefficient of the out-of-sample average return is 36%; if only factors with t-statistics between 2.0 and 5.0 are considered, this coefficient increases to 53%. (As a comparison, in my empirical study using 95 factors in the Chinese stock market, it was found that the shrinkage coefficient is consistent with the results of Harvey and Liu (2021b) for the U.S. stock market.)

With the simulated publication process, the Harvey and Liu (2021b) model can make inferences about the published factors, and the results are particularly valuable. Specifically, they considered the false publication rate and the false non-publication rate. The former is defined as the proportion of false factors that are incorrectly published; the latter is defined as the proportion of genuine factors that are not published. Under the $K_0 = 5000$ parameter set, the former is 1.12%, and the latter is 72.93%. At first glance, 1.12% seems low, but given $K_0 = 5000$ and $p_0 = 0.06$, the total number of false factors is $(1 - 0.06) \times 5000 = 4700$. Among them, 1.12%, or $1.12\% \times 4700 = 53$ factors, were published. These published false factors led to the p-hacking problem we see today. Identifying these published false factors, rather than claiming that p-hacking alone cannot explain factors with very high t-statistics (it certainly cannot!), is the problem that needs to be addressed.

In addition to Chen (2021), Jensen et al. (2023) is another article that defends the findings of previous empirical research. This paper, through an empirical Bayesian model, finds that even when considering the problem of multiple hypothesis testing, the standard errors of factor average returns do not need to be significantly enlarged (i.e., significance is not notably affected), implying that the vast majority of published factors are valid, and there is no replication crisis in financial empirical research. However, their model also implicitly requires specifying the proportion of genuine factors (i.e., p_0). If the prior assumes that the proportion of genuine factors is sufficiently high, then multiple hypothesis testing indeed does not have a substantial impact, and the traditional 2.0 threshold can still be used to test factors. (Using the 95 factors in the aforementioned empirical study, we found that when p_0 is set to 50% (i.e., assuming half of the factors are true), a 2.0 t-statistic threshold controls the false discovery rate under 4%.) Ultimately, a reasonable prior regarding the proportion of genuine factors is crucial for correctly addressing the issue of multiple hypothesis testing.

Regarding multiple hypothesis testing, other disciplines have long recognized its importance, while finance has recently begun to pay attention to it. The good news is that, after nearly a decade of development, there is now awareness of the problem in finance, and various methods (penalizing multiple hypothesis testing, Bayesian approaches, requiring top journals to publish data and code for result replication, etc.) are being employed to reduce the impact of p-hacking.

As for the severity of this issue, it is essential for the academic community to discuss it with an open mind. Harvey and Liu (2021b)'s analysis shows that since we only observe the published factors and do not know how many factors have actually been tested, this problem is inherently unidentifiable. Because of this, research on p-hacking indeed contains a subjective element. The appropriate research attitude is to candidly acknowledge this systemic econometric issue and derive convincing conclusions through reasonable priors.

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